

Transformations

5.1 Introduction

Almost all graphics systems allow the programmer to define picture that include a variety of transformations. For example, the programmer is able to magnify a picture so that detail appears more clearly, or reduce it so that more of the picture is visible. The programmer is also able to rotate the picture so that he can see it in different angles.

In this chapter we discuss the 2D transformations.

5.2 Two Dimensional Transformations

In this section, we describe the general procedures for applying translation, rotation, and scaling parameters to reposition and resize the two dimensional objects.

5.2.1 Translation

Translation is a process of changing the position of an object in a straight-line path from one coordinate location to another. We can translate a two dimensional point by adding translation distances, t_x and t_y , to the original coordinate position (x, y) to move the point to a new position (x', y') , as shown in the Fig. 5.1.

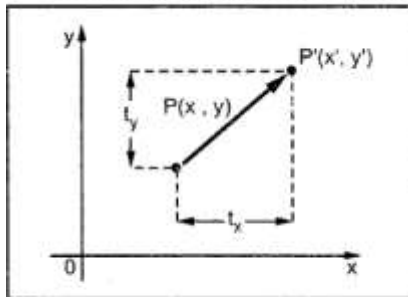


Fig. 5.1

$$x' = x + t_x \quad \dots (5.1)$$

$$y' = y + t_y \quad \dots (5.2)$$

The translation distance pair (t_x, t_y) is called a **translation vector** or **shift vector**.

It is possible to express the translation equations 5.1 and 5.2 as a single matrix equation by using column vectors to represent coordinate positions and the translation vector :

$$P = \begin{bmatrix} x \\ y \end{bmatrix} \quad P' = \begin{bmatrix} x' \\ y' \end{bmatrix} \quad T = \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

This allows us to write the two dimensional translation equations in the matrix form :

$$P' = P + T \quad \dots (5.3)$$

➡ **Example 4.1 :** Translate a polygon with co-ordinates A(2, 5), B(7, 10) and C(10, 2) by 3 units in x direction and 4 units in y direction.

Solution :

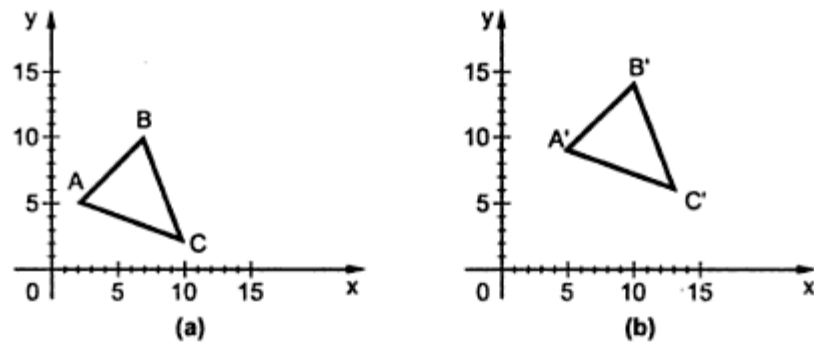


Fig. 4.5 Translation of polygon

$$\begin{aligned} A' &= A + T \\ &= \begin{bmatrix} 2 \\ 5 \end{bmatrix} + \begin{bmatrix} 3 \\ 4 \end{bmatrix} \\ &= \begin{bmatrix} 5 \\ 9 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} B' &= B + T \\ &= \begin{bmatrix} 7 \\ 10 \end{bmatrix} + \begin{bmatrix} 3 \\ 4 \end{bmatrix} \\ &= \begin{bmatrix} 10 \\ 14 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} C' &= C + T \\ &= \begin{bmatrix} 10 \\ 2 \end{bmatrix} + \begin{bmatrix} 3 \\ 4 \end{bmatrix} \\ &= \begin{bmatrix} 13 \\ 6 \end{bmatrix} \end{aligned}$$

4.3.2 Rotation

A two dimensional rotation is applied to an object by repositioning it along a circular path in the xy plane. To generate a rotation, we specify a rotation angle θ and the position of the rotation point about which the object is to be rotated.

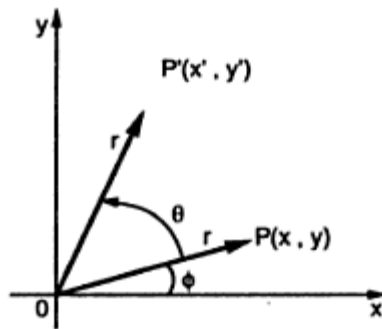


Fig. 4.6

Let us consider the rotation of the object about the origin, as shown in the Fig. 4.6.

Here, r is the constant distance of the point from the origin, angle ϕ is the original angular position of the point from the horizontal and θ is the rotation angle. Using standard trigonometric equations, we can express the transformed co-ordinates in terms of angles θ and ϕ as

$$\left. \begin{aligned} x' &= r \cos(\phi + \theta) = r \cos \phi \cos \theta - r \sin \phi \sin \theta \\ y' &= r \sin(\phi + \theta) = r \cos \phi \sin \theta + r \sin \phi \cos \theta \end{aligned} \right\} \quad \dots(4.4)$$

The original co-ordinates of the point in polar co-ordinates are given as,

$$\left. \begin{aligned} x &= r \cos \phi \\ y &= r \sin \phi \end{aligned} \right\} \quad \dots(4.5)$$

Substituting equations (4.5) into (4.4), we get the transformation equations for rotating a point (x, y) through an angle θ about the origin as :

$$\left. \begin{aligned} x' &= x \cos \theta - y \sin \theta \\ y' &= x \sin \theta + y \cos \theta \end{aligned} \right\} \quad \dots (4.6)$$

The above equations can be represented in the matrix form as given below.

$$[x' \ y'] = [x \ y] \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

$$\therefore P' = P \cdot R \quad \dots (4.7)$$

where R is rotation matrix and it is given as,

$$R = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \quad \dots (4.8)$$

It is important to note that positive values for the rotation angle define counterclockwise rotations about the rotation point and negative values rotate objects in the clockwise sense.

For negative values of θ i.e., for clockwise rotation, the rotation matrix becomes

$$\begin{aligned} R &= \begin{bmatrix} \cos(-\theta) & \sin(-\theta) \\ -\sin(-\theta) & \cos(-\theta) \end{bmatrix} \\ &= \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \quad \because \cos(-\theta) = \cos \theta \text{ and } \sin(-\theta) = -\sin \theta \end{aligned} \quad \dots (4.9)$$

➡ **Example 4.2 :** A point (4, 3) is rotated counterclockwise by an angle of 45° . Find the rotation matrix and the resultant point.

Solution :

$$R = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} = \begin{bmatrix} \cos 45^\circ & \sin 45^\circ \\ -\sin 45^\circ & \cos 45^\circ \end{bmatrix}$$

$$= \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

$$\therefore P' = [4 \ 3] \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

$$= [4/\sqrt{2} - 3/\sqrt{2} \quad 4/\sqrt{2} + 3/\sqrt{2}]$$

$$= [1/\sqrt{2} \quad 7/\sqrt{2}]$$

4.3.3 Scaling

A scaling transformation changes the size of an object. This operation can be carried out for polygons by multiplying the co-ordinate values (x, y) of each vertex by scaling factors S_x and S_y to produce the transformed co-ordinates (x', y').

$$x' = x \cdot S_x$$

$$\text{and} \quad y' = y \cdot S_y \quad \dots (4.10)$$

Scaling factor S_x scales object in the x direction and scaling factor S_y scales object in the y direction. The equation (4.10) can be written in the matrix form as given below :

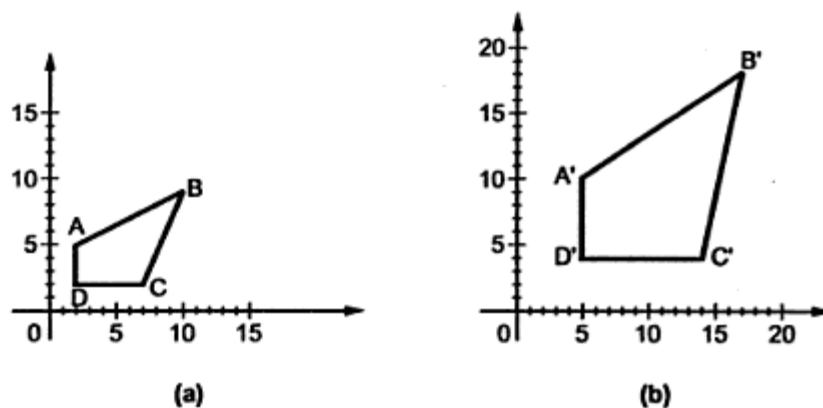


Fig. 4.7

$$[x' \ y'] = [x \ y] \begin{bmatrix} S_x & 0 \\ 0 & S_y \end{bmatrix}$$

$$= P.S$$

Any positive numeric values are valid for scaling factors S_x and S_y . Values less than 1 reduce the size of the objects and values greater than 1 produce an enlarged object. For both S_x and S_y values equal to 1, the size of object does not change. To get uniform scaling it is necessary to assign same value for S_x and S_y . Unequal values for S_x and S_y result in a differential scaling.

Ex. 5.3 : Scale the polygon with coordinates A (2, 5), B (7, 10) and C (10, 2) by two units in x direction and two units in y direction.

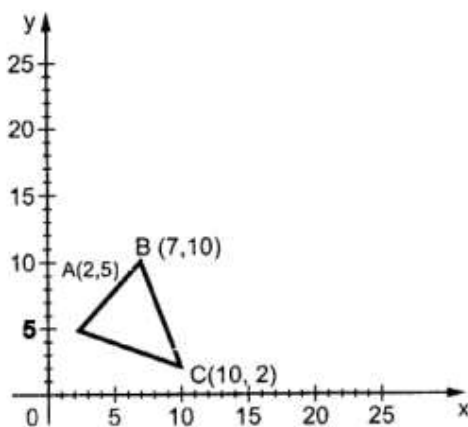
Sol. : Here $S_x = 2$ and $S_y = 2$. Therefore, transformation matrix is given as

$$S = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

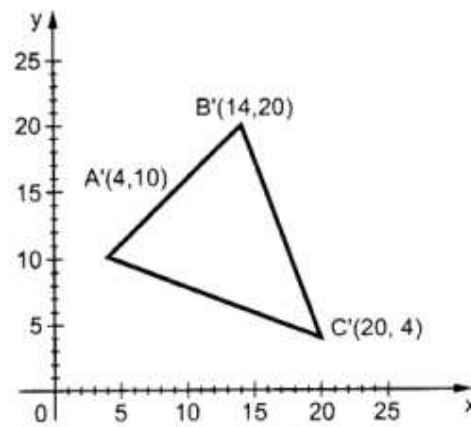
The object matrix is :

$$\begin{array}{c} x \quad y \\ A \begin{bmatrix} 2 & 5 \end{bmatrix} \\ B \begin{bmatrix} 7 & 10 \end{bmatrix} \\ C \begin{bmatrix} 10 & 2 \end{bmatrix} \end{array}$$

$$\therefore \begin{array}{c} A' \begin{bmatrix} x'_1 & y'_1 \end{bmatrix} \\ B' \begin{bmatrix} x'_2 & y'_2 \end{bmatrix} \\ C' \begin{bmatrix} x'_3 & y'_3 \end{bmatrix} \end{array} = \begin{array}{c} \begin{bmatrix} 2 & 5 \\ 7 & 10 \\ 10 & 2 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \\ \\ = \begin{bmatrix} 4 & 10 \\ 14 & 20 \\ 20 & 4 \end{bmatrix} \end{array}$$



(a) Original object



(b) Scaled object

Fig. 5.5

5.3 Homogeneous Coordinates

In design and picture formation process, many times we may require to perform translation, rotations, and scaling to fit the picture components into their proper positions. In the previous section we have seen that each of the basic transformations can be expressed in the general matrix form

$$P' = P \cdot M_1 + M_2 \quad \dots (5.12)$$

For translation :

$$P' = P \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

This part comes in need homogeneous coordinate system

i.e. M_1 = Identity matrix

M_2 = Translation vector

For rotation :

$$P' = P \cdot \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

i.e. M_1 = Rotational matrix

$M_2 = 0$

For scaling :

$$P' = P \cdot \begin{bmatrix} S_x & 0 \\ 0 & S_y \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

i.e. M_1 = Scaling matrix

$M_2 = 0$

To produce a sequence of transformations with above equations, such as translation followed by rotation and then scaling, we must calculate the transformed coordinates one step at a time. First, coordinates are translated, then these translated coordinates are scaled, and finally, the scaled coordinates are rotated. But this sequential transformation process is not efficient. A more efficient approach is to combine sequence of transformations into one transformation so that the final coordinate positions are obtained directly from initial coordinates. This eliminates the calculation of intermediate coordinate values.

In order to combine sequence of transformations we have to eliminate the matrix addition associated with the translation terms in M_2 (Refer equation 5.12). To achieve this we have to represent matrix M_1 as 3×3 matrix instead of 2×2 introducing an additional dummy coordinate W . Here, points are specified by three numbers instead of two. This coordinate system is called **homogeneous coordinate system** and it allows us to express all transformation equations as matrix multiplication.

The homogeneous coordinate is represented by a triplet (X_w, Y_w, W) ,

where

$$x = \frac{X_w}{W} \quad \text{and} \quad y = \frac{Y_w}{W}$$

For two dimensional transformations, we can have the homogeneous parameter W to be any non zero value. But it is convenient to have $W = 1$. Therefore, each two dimensional position can be represented with homogeneous coordinate as $(x, y, 1)$.

Summarizing it all up, we can say that the homogeneous coordinates allow combined transformation, eliminating the calculation of intermediate coordinate values and thus save required time for transformation and memory required to store the intermediate coordinate values. Let us see the homogeneous coordinates for three basic transformations.

5.3.1 Homogeneous Coordinates for Translation

The homogeneous coordinates for translation are given as

$$T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ t_x & t_y & 1 \end{bmatrix} \quad \dots (5.13)$$

Therefore, we have

$$\begin{aligned} [x' \ y' \ 1] &= [x \ y \ 1] \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ t_x & t_y & 1 \end{bmatrix} \\ &= [x + t_x \ y + t_y \ 1] \quad \dots (5.14) \end{aligned}$$

5.3.2 Homogeneous Coordinates for Rotation

The homogeneous coordinates for rotation are given as

$$R = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \dots (5.15)$$

Therefore, we have

$$\begin{aligned} [x' \ y' \ 1] &= [x \ y \ 1] \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= [x \cos\theta - y \sin\theta \quad x \sin\theta + y \cos\theta \quad 1] \quad \dots (5.16) \end{aligned}$$

5.3.3 Homogeneous Coordinates for Scaling

The homogeneous coordinate for scaling are given as

$$S = \begin{bmatrix} S_x & 0 & 0 \\ 0 & S_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Therefore, we have

$$\begin{aligned} [x' \ y' \ 1] &= [x \ y \ 1] \begin{bmatrix} S_x & 0 & 0 \\ 0 & S_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= [x \cdot S_x \ y \cdot S_y \ 1] \end{aligned} \quad \dots (5.17)$$

Note : In this text, the object matrix is written first and it is then multiplied by the required transformation matrix. If we wish to write the transformation matrix first and then the object matrix we have to take the transpose of both the matrices and post-multiply the object matrix i.e.,

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Ex. 5.4 : Give a 3×3 homogeneous coordinate transformation matrix for each of the following translations

- Shift the image to the right 3-units
- Shift the image up 2 units
- Move the image down $\frac{1}{2}$ unit and right 1 unit
- Move the image down $\frac{2}{3}$ unit and left 4 units

Sol. : We know that homogenous coordinates for translation are

$$T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ t_x & t_y & 1 \end{bmatrix}$$

a) Here, $t_x = 3$ and $t_y = 0$

$$\therefore T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix}$$

b) Here, $t_x = 0$ and $t_y = 2$

$$\therefore T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix}$$

c) Here, $t_x = 1$ and $t_y = -0.5$

$$\therefore T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & -0.5 & 1 \end{bmatrix}$$

d) Here, $t_x = -4$ and $t_y = -0.66$

$$\therefore T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & -0.66 & 1 \end{bmatrix}$$

Ex. 4.5 : Find the transformation matrix that transforms the given square ABCD to half its size with centre still remaining at the same position. The coordinates of the square are : A(1, 1), B(3, 1), C(3, 3), D(1, 3) and centre at (2, 2). Also find the resultant coordinates of square.

Sol. : This transformation can be carried out in the following steps.

1. Translate the square so that its center coincides with the origin.
2. Scale the square with respect to the origin.
3. Translate the square back to the original position.

Thus, the overall transformation matrix is formed by multiplication of three matrices.

$$\begin{aligned} \therefore T_1 \cdot S \cdot T &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & -2 & 1 \end{bmatrix} \begin{bmatrix} 0.5 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 2 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 0.5 & 0 & 0 \\ 0 & 0.5 & 0 \\ -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 2 & 1 \end{bmatrix} \end{aligned}$$

$$\begin{aligned}
&= \begin{bmatrix} 0.5 & 0 & 0 \\ 0 & 0.5 & 0 \\ 1 & 1 & 1 \end{bmatrix} \\
\begin{bmatrix} A' \\ B' \\ C' \\ D' \end{bmatrix} &= \begin{bmatrix} 1 & 1 & 1 \\ 3 & 1 & 1 \\ 3 & 3 & 1 \\ 1 & 3 & 1 \end{bmatrix} \begin{bmatrix} 0.5 & 0 & 0 \\ 0 & 0.5 & 0 \\ 1 & 1 & 1 \end{bmatrix} \\
&= \begin{bmatrix} 1.5 & 1.5 & 1 \\ 2.5 & 1.5 & 1 \\ 2.5 & 2.5 & 1 \\ 1.5 & 2.5 & 1 \end{bmatrix}
\end{aligned}$$

5.4 Composition of 2D Transformations

We have seen what is meant by combined or concatenated or composed transformations in the previous section. The basic purpose of composing transformations is to gain efficiency by applying a single composed transformation to a point, rather than applying a series of transformations, one after the other.

5.4.1 Rotation About an Arbitrary Point

To rotate an object about an arbitrary point, (x_p, y_p) we have to carry out three steps :

1. Translate point (x_p, y_p) to the origin
2. Rotate it about the origin and
3. Finally, translate the center of rotation back where it belongs (See Fig.5.6)

We have already seen that matrix multiplication is not commutative, i.e. multiplying matrix A by matrix B will not always yield the same result as multiplying matrix B by matrix A. Therefore, in obtaining composite transformation matrix, we must be careful to order the matrices so that they correspond to the order of the transformations on the object. Let us find the transformation matrices to carry out individual steps.

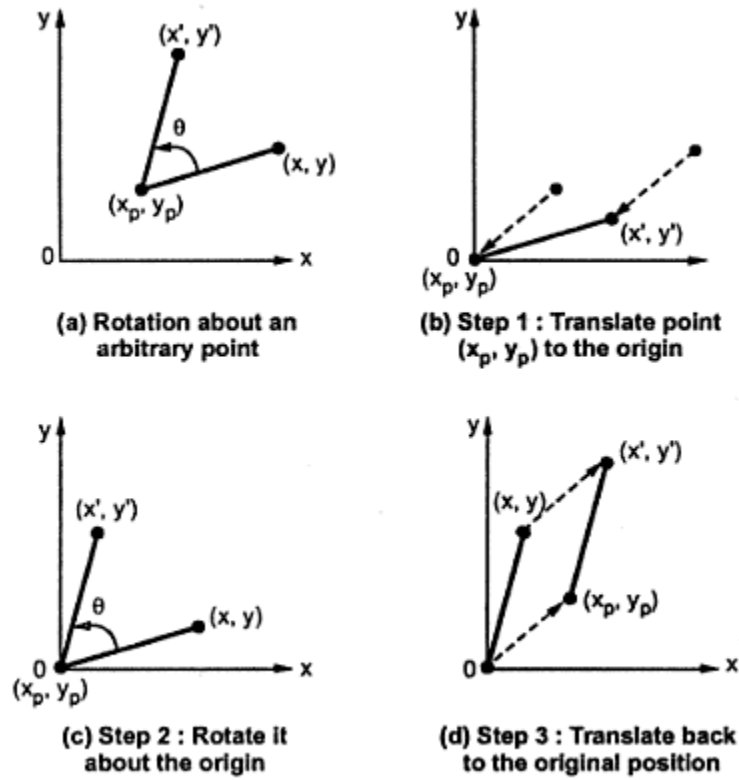


Fig. 4.9

The translation matrix to move point (x_p, y_p) to the origin is given as,

$$T_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -x_p & -y_p & 1 \end{bmatrix}$$

The rotation matrix for counterclockwise rotation of point about the origin is given as,

$$R = \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The translation matrix to move the center point back to its original position is given as,

$$T_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ x_p & y_p & 1 \end{bmatrix}$$

Therefore, the overall transformation matrix for a counterclockwise rotation by an angle θ about the point (x_p, y_p) is given as,

$$\begin{aligned}
 T_1 \cdot R \cdot T_2 &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -x_p & -y_p & 1 \end{bmatrix} \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ x_p & y_p & 1 \end{bmatrix} \\
 &= \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ -x_p \cos \theta + y_p \sin \theta & -x_p \sin \theta - y_p \cos \theta & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ x_p & y_p & 1 \end{bmatrix} \\
 &= \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ -x_p \cos \theta + y_p \sin \theta + x_p & -x_p \sin \theta - y_p \cos \theta + y_p & 1 \end{bmatrix} \dots (4.18)
 \end{aligned}$$

►► **Example 4.7 :** Perform a counterclockwise 45° rotation of triangle $A(2, 3)$, $B(5, 5)$, $C(4, 3)$ about point $(1, 1)$.

Solution : From equation (4.18) we have,

$$T_1 \cdot R \cdot T_2 = \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ -x_p \cos \theta + y_p \sin \theta + x_p & -x_p \sin \theta - y_p \cos \theta + y_p & 1 \end{bmatrix}$$

Here, $\theta = 45^\circ$, $x_p = 1$ and $y_p = 1$. Substituting values we get,

$$T_1 \cdot R \cdot T_2 = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ -1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 1 & -\sqrt{2}+1 & 1 \end{bmatrix}$$

$$\therefore \begin{bmatrix} A' \\ B' \\ C' \end{bmatrix} = \begin{bmatrix} 2 & 3 & 1 \\ 5 & 5 & 1 \\ 4 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ -1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 1 & -\sqrt{2}+1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{1}{\sqrt{2}}+1 & \frac{3}{\sqrt{2}}+1 & 1 \\ 1 & \frac{8}{\sqrt{2}}+1 & 1 \\ \frac{1}{\sqrt{2}}+1 & \frac{5}{\sqrt{2}}+1 & 1 \end{bmatrix}$$

4.5.2 Shear

A transformation that slants the shape of an object is called the shear transformation. Two common shearing transformations are used. One shifts x coordinate values and other shifts y coordinate values. However, in both the cases only one coordinate (x or y) changes its coordinates and other preserves its values.

4.5.2.1 X shear

The x shear preserves the y coordinates, but changes the x values which causes vertical lines to tilt right or left as shown in the Fig. 4.8. The transformation matrix for x shear is given as

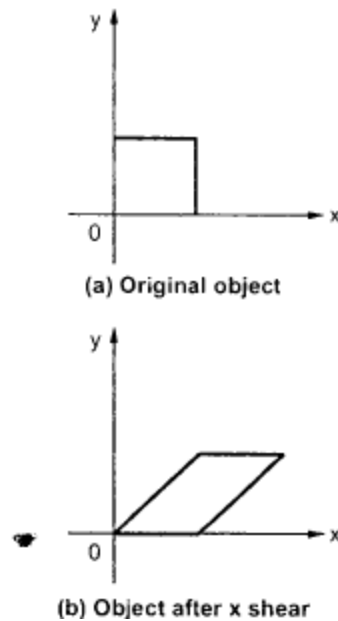


Fig. 4.8

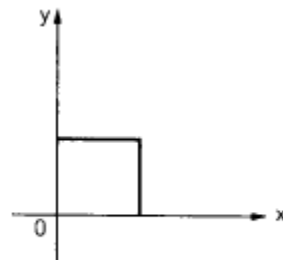
$$X_{sh} = \begin{bmatrix} 1 & 0 & 0 \\ Sh_x & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore x' = x + Sh_x \cdot y \quad \text{and}$$

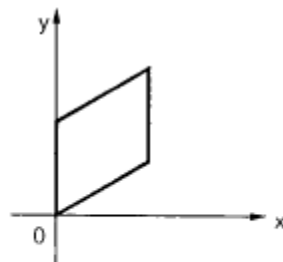
$$y' = y \quad \dots (4.19)$$

4.5.2.2 Y shear

The y shear preserves the x coordinates, but changes the y values which causes horizontal lines to transform into lines which slope up or down, as shown in the Fig. 4.9.



(a) Original object



(b) Object after y shear

Fig. 4.9

The transformation matrix for y shear is given as

$$Y_{sh} = \begin{bmatrix} 1 & Sh_y & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore x' = x \text{ and } y' = y + Sh \cdot x$$

5.5 Other Transformations

The three basic transformations of scaling, rotating, and translating are the most useful and most common. There are some other transformations which are useful in certain applications. Two such transformations are reflection and shear.

5.5.1 Reflection

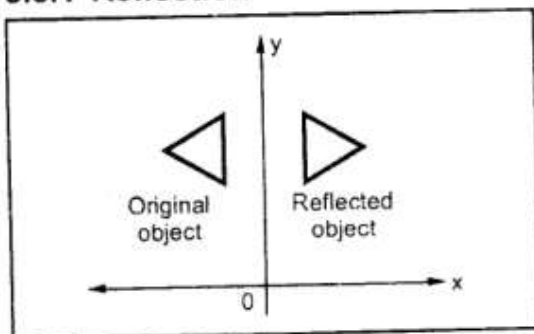


Fig. 5.7 Reflection about y axis

A reflection is a transformation that produces a mirror image of an object relative to an axis of reflection. We can choose an axis of reflection in the xy plane or perpendicular to the xy plane. The table 5.1 gives examples of some common reflections.

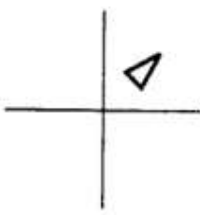
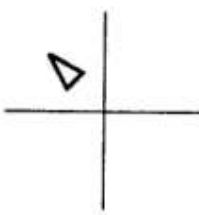
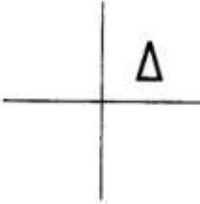
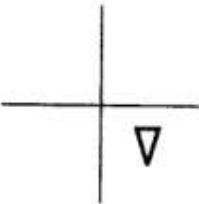
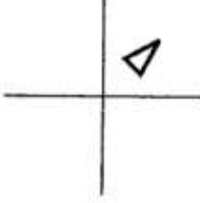
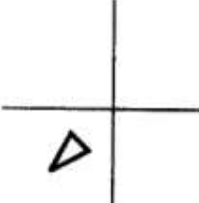
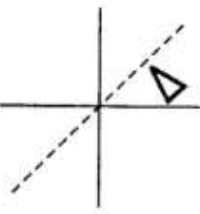
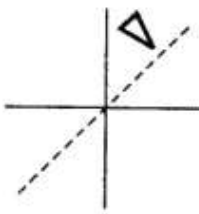
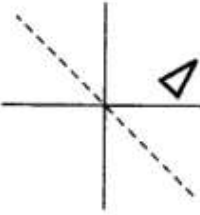
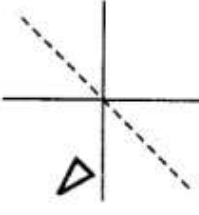
Reflection	Transformation matrix	Original image	Reflected image
Reflection about Y-axis	$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$		
Reflection about X axis	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$		
Reflection about origin	$\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$		
Reflection about line $y = x$	$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$		
Reflection about line $y = -x$	$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$		

Table 5.1 Common reflections

